Applied Causal Inference: A Gentle Stroll Down the Rabbit Hole

PhD Programme in Economics, Management and Methods
for the Sustainable Transition

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Course Description

This short course introduces the main quasi-experimental methods used in empirical economics. The goal is to give PhD students a working knowledge of each method: the identifying assumptions, how to implement them in Stata, and how to interpret and communicate results. Each lecture combines theory with a hands-on Stata lab using published replication data.

The course assumes familiarity with OLS, panel data, and basic probability theory. No prior exposure to causal inference is required.

Schedule

Date	Time	Lecture	Total
5 May	09:30–13:00	*Counterfactual Framework and Methods Based on Selection on Observables: 'Cause We Are Living in a Counterfactual World*	3.5h (2h+1.5h)
5–6 May	14:00–16:00 / 09:30–16:00	*Difference-in-Differences: the Good, the Bad and the Ugly*	8h (4h+4h)
7 May	09:00–13:00	*Of Synthetic Realms, Discontinuities and Shift-Shares or: How I Learned to Stop Worrying and Love the Process*	4h (2h+2h)

Theory + lab hours in parentheses. All sessions at the University of Ferrara.

1. *Counterfactual Framework and Methods Based on Selection on Observables: 'Cause We Are Living in a Counterfactual World* (5 May, morning)

Theory

- Motivation: why correlation is not causation; the fundamental problem
- The potential outcomes framework (Rubin, 1980): $Y_i(1)$, $Y_i(0)$, ATE, ATT, LATE

- Selection bias and the conditional independence assumption (CIA)
- From description to causation: identification strategies
- Estimators for selection on observables: matching (nearest-neighbour, kernel, propensity score) and inverse probability weighting (IPW)
- Overlap, common support, and the limits of observational inference

Lab: Titanic and LaLonde

Two exercises. First, the Titanic passenger data to build intuition about selection bias and potential outcomes. Second, LaLonde (1986)'s job training programme (NSW): students compare experimental and observational estimates of the training effect on earnings using matching and IPW.

2. *Difference-in-Differences: the Good, the Bad and the Ugly* (5 May PM + 6 May)

Theory

- The 2×2 DiD: logic, parallel trends, and the estimator
- Controls in DiD: when adding covariates helps and when it hurts
- Two-way fixed effects (TWFE): the workhorse and its assumptions
- Staggered adoption: why TWFE breaks down with heterogeneous timing
- Modern estimators: Callaway and Sant'Anna (2021) (CS), Sun and Abraham (2021) (SA), de Chaisemartin and D'Haultfœuille (2020)
- Event-study designs, pre-trend testing, and what they can and cannot show
- Sensitivity analysis: relaxing parallel trends (Rambachan and Roth, 2023)

Lab: Two applications

Exercise 1 — Rizzo et al. (2025), Departments of Excellence. Rizzo et al. (2025): effect of the Italian Academic Excellence Initiative on faculty recruitment (290 departments, 2013–2020). Main DiD estimate, event study, and CS estimator.

Exercise 2 — Rentocchini et al. (2025), Superstar M&A. Rentocchini (2025): do tech acquisitions raise firm-level markups? TWFE and de Chaisemartin & D'Haultfœuille estimator.

3. *Of Synthetic Realms, Discontinuities and Shift-Shares or: How I Learned to Stop Worrying and Love the Process* (7 May)

Theory

- Synthetic Control (Abadie et al., 2010): data-driven counterfactual for aggregate units; convex-hull identification; permutation inference

- Sharp RDD (Cattaneo et al., 2019): local randomisation near a cutoff; bandwidth selection; RD plots; validity checks
- Shift-Share IV (Bartik, 1991): the Bartik instrument $z_\ell = \sum_n s_{\ell n} \cdot g_n$; shock vs. share exogeneity (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020); Rotemberg weights

Lab: Three applications

Meyersson (2014) (RDD, Turkish elections, female education), Pinotti (2015) (Synthetic Control, mafia and GDP), and Autor et al. (2013) (Shift-Share IV, the China Shock).

Software

All lab sessions use **Stata 19** (but should work in previous versions as well). Replication data and do-files are available at `https://github.com/frarent/fit_phd_lectures`. Open the corresponding stata project file (`e.g. 00_lab1.stpr` within each lecture's `Lab/LN/` directory; required packages are bundled in the repository (no need to install packages).

References

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.

Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, 103(6):2121–2168.

Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* W.E. Upjohn Institute for Employment Research, Kalamazoo, MI.

Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1):181–213.

Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

Cattaneo, M. D., Idrobo, N., and Titiunik, R. (2019). *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge University Press.

de Chaisemartin, C. and D'Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.

Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.

LaLonde, R. J. (1986). Evaluating the econometric evaluations of training programs with experimental data. *American Economic Review*, 76(4):604–620.

Meyersson, E. (2014). Islamic rule and the empowerment of the poor and pious. *Econometrica*, 82(1):229–269.

Pinotti, P. (2015). The economic costs of organised crime: Evidence from southern Italy. *The Economic Journal*, 125(586):F203–F232.

Rambachan, A. and Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591.

Rentocchini, F. (2025). Superstar m&a: Technology acquisitions and firm performance. Working paper.

Rizzo, U., Rentocchini, F., Seeber, M., and Ramaciotti, L. (2025). Beyond the badge of honour: The effect of the academic excellence initiative on staff recruitment in italy. *Oxford Economic Papers*. Forthcoming.

Rubin, D. B. (1980). Randomization analysis of experimental data: The fisher randomization test comment. *Journal of the American Statistical Association*, 75(371):591–593.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.